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Decision & Branching Structure

Programming Logic · Week 7 · 28 October 2026

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Learning Outcomes

By the end of this session, you can control which path a program takes based on a condition.

Explain

Why programs decide and how a condition controls flow

Write

Conditions with relational and logical operators

Design

Flowcharts and pseudocode for IF, ELSE IF, nested IF

```
INPUT score
IF score >= 60 THEN
    OUTPUT "Pass"
ELSE
    OUTPUT "Fail"
END IF
```

IF and IF-ELSE

A condition is TRUE or FALSE. IF runs a block only when true. IF-ELSE picks one of two paths, never both.

Building Conditions

Relational

Symbol	Meaning
==	equal
!=	not equal
>	greater
<	less
>=	at least
<=	at most

Logical

Logic	True when
AND	both true
OR	any true
NOT	reverses
1st	NOT
2nd	AND
3rd	OR

Examples (x = 15)

Condition	Result
x > 10	True
x == 20	False
x>10 AND x<20	True
x<5 OR x>12	True
NOT (x == 15)	False
x != 15	False

```
IF score >= 80 THEN grade = "A"
ELSE IF score >= 70 THEN grade = "B"
ELSE IF score >= 60 THEN grade = "C"
ELSE grade = "D"
END IF
```

ELSE IF Ladder

STEP 01

Test first

Check the top condition. If true, run it and skip the rest.

STEP 02

Test next

If false, move down to the next condition.

STEP 03

Else

If none is true, the final ELSE runs.

```
IF hasTicket THEN
    IF age >= 17 THEN
        OUTPUT "Enter"
    ELSE
        OUTPUT "Need guardian"
    END IF
ELSE
    OUTPUT "Buy a ticket"
END IF
```

Nested IF

A decision inside a decision. The inner IF is checked only when the outer condition is true.

Common Logic Mistakes

Mistake 1

Wrong order

Test the narrowest case first, or every A falls into C.

Mistake 2

AND vs OR

Between 10 and 20 needs AND. OR accepts every number.

Mistake 3

Boundary case

Decide what happens at exactly 60, not only above or below.

Mistake 4

= vs ==

= assigns a value. == compares two values.

Mistake 5

No final ELSE

Without it, unexpected input is silently ignored.

Mistake 6

Too deep

Deep nesting hides bugs. Flatten with AND or ELSE IF.

Ticket Price Problem

Input a person's age and whether the visit is on a weekend.

Ticket price

- Under 12: 20,000
- 12 to 59: 50,000
- 60 and above: 30,000
- Weekend: add 10,000 to every ticket

Tasks

- Draw the flowchart
- Write the pseudocode
- Trace the program with 3 test cases

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